

Reminder: Empirical and simulation exercises must be submitted with “stand-alone” code and data (if any) in an appropriate format. The homework is due Sunday (data, code, and word-processed report by email to **the instructor before 10 PM Sunday**). Also, submit your work-processed report via **Gradescope**.

Consider the same model from Homework #9.

1. **Bayesian Estimation.** Using the log-likelihood from the Kalman filter, estimate θ using MCMC. That is, $p(\theta) = 1$ (uninformative prior), and we effectively perform maximum likelihood estimation by finding the posterior mode using MCMC. Make sure that the code for evaluating the likelihood in the Kalman filter is fast (you can use Matlab’s command “profile” to identify bottlenecks).
 - (a) Set initial value for θ at the true parameter values.
 - (b) Using the Metropolis-Hastings algorithm, generate the MCMC chain where you draw each parameter θ separately and the objective function is the log-likelihood from the Kalman filter for a given θ . In other words, draw candidate value of θ as follows

$$\vartheta = \theta^{(i-1)} + c N(0, \hat{\Sigma}),$$
 where $N(0, \hat{\Sigma})$ is a 3-dimensional random normal vector and $\hat{\Sigma}$ is the negative inverse of Hessian obtained in 1(c). Keep track of accepted values. Start from $c = 1$. Adjust c every 1,500 draws by 10 percent up if the accept rate for a given parameter is above 40% and by 10 percent down if accept rate for a given parameter is below 25%. Adjust the size of the shocks during the first 30,000 draws. Keep c fixed after the first 30,000 draws. The size of the generated chain should be 200,000 draws.
 - (c) Plot the generated chains for each parameter. Does the chain for each parameter converge to a stable distribution?
 - (d) Plot the path of the value of the objective function corresponding to the generated path of the parameter values in the MCMC chain. Does the value of the objective function stabilize after a large number of draws (you may have to drop the first 1,000 or so draws to see a clear picture)?
 - (e) Plot the scatter plot of the generated parameter values in the chain against the generated path of the values of the objective function. How can you interpret these scatter plots?

HINT: See the sample example code available on Canvas, which illustrates how to estimate the following regression model using a quasi-Bayesian method based on MCMC:

$$\begin{aligned} y_t &= \alpha x_t + \alpha^2 z_t + e_t, & \alpha &= 2, \\ x_t &\sim N(1, 1^2), & z_t &\sim N(1, 1^2), \\ e_t &\sim N(0, 10^2), & t &= 1, \dots, 100. \end{aligned}$$